



LAB ASSIGNMENT -4

Object oriented programming

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Course – MCA (26-28)

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1. Create a C++ program to demonstrate Single Inheritance.

Create a base class Person with the following data members:

Name

Age

Create a derived class Student having:

Roll Number

Course

Implement member functions to accept and display the complete details of a student using inheritance.

CODE

```
#include <iostream>
using namespace std;
class Person {
protected: string
name; int age;

public: void getPersonDetails
() { cout << "Enter Name: ";
getline(cin, name);

        cout << "Enter Age: ";
        cin >> age; cin.ignore
        ();
    }
};

class Student : public Person {
private: int rollNumber; string
course;

public: void
acceptDetails() {
getPersonDetails();

        cout << "Enter Roll Number: ";
        cin >> rollNumber; cin.ignore
        ();

        cout << "Enter Course: ";
        getline(cin, course);
    }
}
```

```

void displayDetails() { cout << "\n---
Student Details ---\n"; cout << "Name:
" << name << endl; cout << "Age: " << age <<
endl; cout << "Roll Number: " << rollNumber <<
endl; cout << "Course: " << course << endl;

}
};

int main() {
    Student s;

    s.acceptDetails();
    s.displayDetails();

    return 0;
}

```

```

Enter Name: student1
Enter Age: 4
Enter Roll Number: 2411
Enter Course: MCA

--- Student Details ---
Name: student1
Age: 4
Roll Number: 2411
Course: MCA

...Program finished with exit code 0
Press ENTER to exit console.

```

2. Create a C++ program to demonstrate Multilevel Inheritance.

Create three classes:

Person → stores Name

Employee → stores Employee ID

Manager → stores Department

Derive Employee from Person and Manager from Employee. Display all the details of the manager.

CODE

```

#include <iostream>
using namespace std;

class Person {
protected:
string name;

```

```
public: void getName() { cout  
<< "Enter Name: "; getline  
(cin, name);
```

```
}};
```

```
class Employee : public Person {  
protected:  
    int employeeID;
```

```
public: void  
getEmployeeDetails() {  
    getName();  
  
        cout << "Enter Employee ID: ";  
        cin >> employeeID; cin.ignore  
        (); }
```

```
};
```

```
class Manager : public Employee {  
private:  
    string department;
```

```
public: void acceptDetails()  
{ getEmployeeDetails();
```

```
  
        cout << "Enter Department: ";  
        getline(cin, department);  
    }
```

```
void displayDetails() { cout << "\n---  
Manager Details ---\n"; cout << "Name:  
" << name << endl; cout << "Employee ID: " <<  
employeeID << endl; cout << "Department: " <<  
department << endl;  
} };
```

```
int main() {  
    Manager m;
```

```
  
    m.acceptDetails();  
    m.displayDetails();
```

```
    return 0;  
}
```

```
Enter Name: harjan  
Enter Employee ID: 7867  
Enter Department: mca  
  
--- Manager Details ---  
Name: harjan  
Employee ID: 7867  
Department: mca  
  
...Program finished with exit code 0  
Press ENTER to exit console.□
```

3. Create a C++ program to demonstrate Multiple Inheritance.

Create two base classes:

Academic → stores Marks

Sports → stores Sports Score

Create a derived class **Result** that inherits from both classes and calculates the total score and displays the result.

CODE

```
#include <iostream>  
using namespace std;  
  
class Academic {  
protected: float  
marks;  
  
public: void getMarks() { cout << "Enter  
Academic Marks: "; cin >> marks;  
  
}};  
  
class Sports {  
protected: float  
sportsScore;
```

```

public: void getSportsScore() { cout
<< "Enter Sports Score: "; cin >>
sportsScore;

    }
};

class Result : public Academic, public Sports {
private:
    float total;

public: void
calculateResult() {
getMarks();
getSportsScore();

    total = marks + sportsScore; } void displayResult() {
cout << "\n--- Result ---\n"; cout << "Academic
Marks: " << marks << endl; cout << "Sports Score: "
<< sportsScore << endl; cout << "Total Score: " <<
total << endl;

    }
};

int main() {
    Result r;

    r.calculateResult();

    r.displayResult();

    return 0;
}

```

```

Enter Name: harjan
Enter Employee ID: 7867
Enter Department: mca

--- Manager Details ---
Name: harjan
Employee ID: 7867
Department: mca

...Program finished with exit code 0
Press ENTER to exit console.

```

**4. Create a C++ program to demonstrate Hierarchical Inheritance.
Create a base class Vehicle containing:
Brand
Model**

Create two derived classes:

Car → Number of Doors

Bike → Engine Capacity

Create separate objects of Car and Bike and display their respective details.

CODE

```
#include <iostream>
using namespace std;

class Vehicle {
protected:
string brand;
string model;

public: void getVehicleDetails
() { cout << "Enter Brand: ";
getline(cin, brand);

        cout << "Enter Model: ";
        getline(cin, model);
    } };

class Car : public Vehicle {
private: int
numberOfDoors;

public: void getCarDetails() { cout <<
"\nEnter Car Details\n";
getVehicleDetails();

        cout << "Enter Number of Doors: ";
        cin >> numberOfDoors; cin.ignore
        ();
    }

    void displayCarDetails() { cout << "\n---
Car Details ---\n"; cout << "Brand: " <<
brand << endl; cout << "Model: " <<
model << endl;
        cout << "Number of Doors: " << numberOfDoors << endl;
    } };
}
```

```
class Bike : public Vehicle {
private: int engineCapacity;

public: void getBikeDetails() { cout <<
"\nEnter Bike Details\n";
getVehicleDetails();

        cout << "Enter Engine Capacity (CC): ";
        cin >> engineCapacity; cin.ignore();

    }

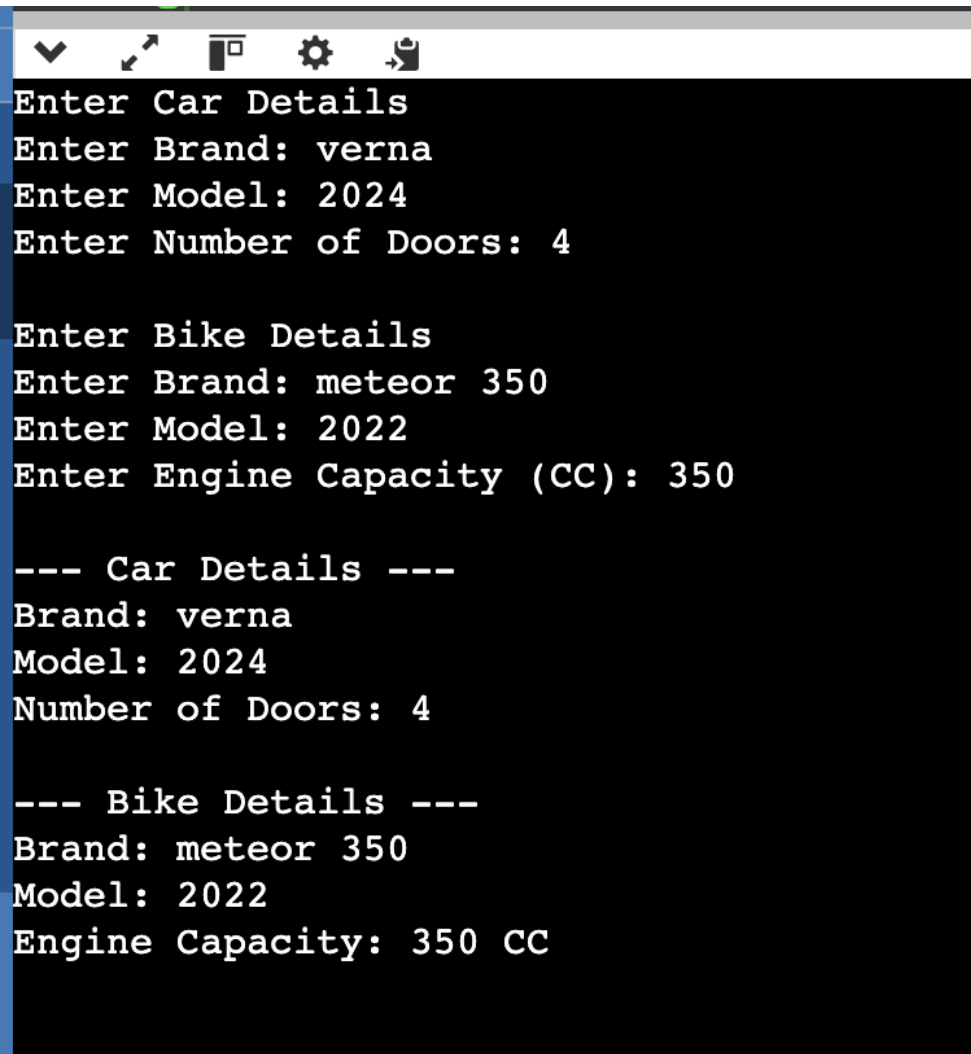
    void displayBikeDetails() { cout <<
"\n--- Bike Details ---\n"; cout <<
"Brand: " << brand << endl; cout <<
"Model: " << model << endl; cout << "Engine Capacity: " <<
engineCapacity << " CC" << endl;
    }
};

int main() {
    Car c;
    Bike b;

    c.getCarDetails();
    b.getBikeDetails();

    c.displayCarDetails();
    b.displayBikeDetails();

    return 0;
}
```

A screenshot of a terminal window with a dark background and light-colored text. The window has a standard OS title bar with icons for window control. The program prompts the user to enter car details (Brand: verna, Model: 2024, Number of Doors: 4) and bike details (Brand: meteor 350, Model: 2022, Engine Capacity: 350 CC). It then displays the entered information in two sections: '--- Car Details ---' and '--- Bike Details ---'.

5 Create a base class College with a constructor that initializes the College Name. Create a derived class Department with a constructor that initializes the Department Name.
Create an object of the derived class and display both college and department information to show the order of constructor execution.

CODE

```
#include <iostream>
using namespace std;

class College {
protected: string
collegeName;

public:
    College(string cName) {
        collegeName = cName;
        cout << "College constructor executed." << endl;
    }
}
```

```

    void displayCollege() { cout << "College Name: " <<
        collegeName << endl;
    }
};

class Department : public College {
private:
    string departmentName;

public:
    Department(string cName, string dName) : College(cName) {
        departmentName = dName; cout << "Department constructor
        executed." << endl;
    }

    void displayDetails() { cout << "\n--- College and Department
        Details ---\n"; displayCollege(); cout << "Department Name: "
        << departmentName << endl;

    }
};

int main() {
    Department d("Lovely Professional University", "Computer Applications");

    d.displayDetails();

    return 0;
}

```

```

College constructor executed.
Department constructor executed.

--- College and Department Details ---
College Name: Lovely Professional University
Department Name: Computer Applications

...Program finished with exit code 0
Press ENTER to exit console.

```